

Two ways to be non-inertial: Frustrativity and event maximality in O'dam

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1. Introduction: frustrative markers

This paper examines two *frustrative* markers in O'dam: *tii* (glossed below as *FRST*) and *tiiip(up)* (*FRST.NONMAX*). Like frustratives in other languages (see Overall 2017), these particles are used to signal *counter-to-expectation* effects: in particular, the non-realization of some intended or expected outcome associated with the proposition described by the marked clause. Despite a recent surge of interest,¹ frustratives remain comparatively understudied: since O'dam is cross-linguistically rare in having two contrastive particles, it provides a unique opportunity to probe the semantic constituents of frustrativity. We argue that the O'dam particles share a version of the counter-to-expectation effect which characterizes frustratives across languages, but that the stronger (more restricted) particle *tiiip(up)* additionally imposes a subclausal *antimaximality* requirement on the instantiation of an event described in the marked clause.

Frustratives combine grammatically with a single clause, but often imply the relevance of a second proposition which acts as the unrealized 'target' of frustration. Their analysis is complicated by variation (within and across languages) in whether this target is a lexically-specified part of the marked clause event or must be recovered pragmatically. Recent work (Copley & Harley 2014, Davis & Matthewson 2022) links this variation to an interaction between frustrativity and grammatical aspect: since frustratives also invoke modal notions of expectation (and/or intention), they evidently operate at the interface between aspect and modality, and their precise range of uses in a given language is sensitive to language-internal resources for marking tense, aspect, and modality. While most accounts take frustratives to be propositional operators, we draw on new data from O'dam to argue that they can also interact with event-structural information. Our analysis thus contributes to the broader typological picture by suggesting that frustratives can draw on resources both at and below the clause level in establishing their core counter-to-expectation effects.

2. Types of frustrativity

A fairly wide range of functions fall under the frustrative umbrella (see Overall 2017). We focus here on a core set which appear to be universal. It is worth noting that frustratives vary crosslinguistically in the extent to which they may invoke unrealized intentions or desires in lieu of expectations: we will discuss O'dam frustratives in terms of expectation, but believe that they are likely in the more permissive class.

Examples (1a)-(1c) exemplify '**proper**' frustrativity (Carol & Salanova 2017). They establish the realization of an event of the marked-clause type, but signal that some associated outcome (often a causal consequence, but see Davis & Matthewson 2022) was not realized. For instance, (1a) conveys that its speaker got bitten despite having shoosed the dog away: frustrative *dara* attaches to the "shooing" description, and the optional second clause spells out a clause-external target of frustration. (1b) illustrates proper frustrativity with an atelic (state) predicate in St'át'imcets, and (1c) shows the same effect with

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¹ For prior work on frustrativity, see: Copley & Harley (2014: Tohono O'dham *cem*), Davis & Matthewson (2022: St'át'imcets *sénaʔ*), Carol & Salanova (2017: Chorote *ta*, Mëbengokre *te*), Kroeger (2017, 2024: Kimaragang *dara*).

a Tohono O’odham accomplishment. Both (1a) and (1b) specify the relevant unexpected effect, but the frustrated target need not be explicit: (1c) can convey the absence of an expected guest, that Juan unexpectedly failed to depart, or even that the door unexpectedly failed to remain open.²

- (1) a. N-o-sii-Ø ku no **dara** it=tasu nga’ n-iit-an oku-i’
 PST-NVOL-shoo-OV 1SG already FRST NOM=dog but PST-bite-DV 1SG=EMPH
 ‘I shooed the dog but I got bitten anyway.’ (Kimaragang, Kroeger 2017)
- b. s-qacw **séna7** ta=n-q’íl’q=a, t’u7 wá7=lhkan=t’u7 ka-mitsa7q-mín=a
 STAT-break FRST DET=1SG.POSS-chair=EXIS but IPFV=1SG.SBJ=EXCL CIRC-sit-RLT-CIRC
 ‘My chair is broken, but I can still sit on it.’ (St’át’imcets, Davis & Matthewson 2022)
- c. Huan ’at **cem** ku:pio g pualt
 Juan aux-PF FRST open DET door
 ‘Juan opened the door in vain.’ (Tohono O’odham, Copley & Harley 2014)

Incompletive frustrativity is usually described in combination with accomplishments in the marked clause. In (2a)–(2b), frustrativity targets the culmination point of the underlying predicate, while establishing that some initial phase of this event was realized. For example, (2a) conveys that Juan acted on the door in a way which would ordinarily lead to its opening, but did not actually get it open. Similarly, (2b) describes a wound that was in the process of healing, but which reopened before the healing was complete. Davis & Matthewson (2022) report similar readings for frustrative-marked accomplishment clauses in St’át’imcets. Crucially, this type of interpretation is not possible with punctual achievements (see 2c), so that incompletive frustrativity appears akin to a strengthened—i.e., *antiresultative*—type of progressivity.

- (2) a. Huan ’o **cem** kukpi’ok g pualt
 Juan aux.IMPF FRST open DET door
 ‘Juan was trying to open the door.’ (Tohono O’odham, Copley & Harley 2014)
- b. k<um>orop no **dara** it=pilat dialo, naka-raa kembagu
 <AV>scab COMPL FRST NOM=wound 3SG PST.AV.NVOL-blood again
 ‘His wound was beginning to heal/form a scab, but then it started bleeding again.’
 (Kimaragang, Kroeger 2024)
- c. #zuqw **séna7** kw=s-Fred, t’u7 ama-wíl’c aylh
 die FRST DET=NMLZ-Fred but good-become now
 ‘Fred died, but he’s okay now.’ (St’át’imcets; Davis & Matthewson 2022)
Comment: “He died, but he’s still alive!”

Avertive frustrativity (3a–3c) conveys that something was going on in the reference context which would plausibly lead to an event of the marked clause type, but that this event did not actually get started. In some languages, the avertive reading seems only to be available in combination with overt futurity (prospectivity) in the marked clause: we see this in both Tohono O’odham (3a) and St’át’imcets (3b).³ This requirement is not crosslinguistically universal, however, since an avertive reading is available for Kimaragang (3c) in the absence of futurate marking (see also Carol & Salanova 2017).

- (3) a. Huan ’at o **cem** kukpi’ok g pualt
 Juan aux.PF FUT FRST open DET door

² This last effect—termination of a result state—is also reported for ‘proper’ uses of St’át’imcets *séna7* with telic predicates (Davis & Matthewson 2022). This is similar to a *cessative* form of frustrativity which arises with state predicates in the marked clause: such claims convey that the state obtained in the past, but does not persist at speech time. As Copley & Harley (2014) point out, cessativity has something in common with proper frustrativity if we assume a default expectation that states (once realized) will continue indefinitely. The observation that examples like (1c) can signal result-state cessation, together with the fact that stative examples like (1b) allow the marked-clause state to persist at speech time further supports classifying proper and cessative frustrativity together.

³ Despite the FUT gloss in (3a), Copley & Harley (2014) treat *o* as a prospective (cf. Zepeda 1983). See Davis & Matthewson (2022) for discussion of the contrastive interaction of frustratives with prospectives vs. modal futures.

‘Juan was going to open the door.’ (but he tripped on the way)

(Tohono O’odham, Copley & Harley 2014)

- b. cúz’=lhkan **séna7** áz’-en na=káoh=a, t’u7 plán=tu7 wa7 lhég•gep
 PROSP=1PL.SBJ FRST buy-DIR DET=car=EXIS but already=DIST IMPF get.sold•FRE
 ‘I was going to buy the car, but it had already been sold.’

(St’át’imcets, Davis & Matthewson 2022)

- c. iit-an oku no **dara** da-tasu nga’ a=tanak po=ot nokoponii
 bite-DV 1SG COMPL FRST GEN=dog but NOM=child FOC=NOM AV.PST.say.sii
 ‘I was about to be bitten by the dog, but the child said “Shii!” (I did not get bitten)

(Kimaragang, Kroeger 2024)

O’dam frustrativity shows all of the above uses, but—as we will see—the two particles do not occupy the functional space uniformly. We also find evidence of several other crosslinguistically common frustrative functions—as indicators of optativity, conditional counterfactuality, and as politeness markers in requests—but set these aside here for reasons of space (see Nadathur & Everdell 2025).

3. Frustrativity in O’dam

O’dam (ISO 639-3 stp)⁴ is a Uto-Aztecan language (Tepiman <) spoken in the Mexican states of Durango, Nayarit, and Zacatecas (see García Salido & Everdell 2020). The language is part of the Southern Tepehuan sub-branch with sister languages Au’dam (Southwestern Tepehuan) and Central Au’dam. The Southern Tepehuan languages have ca.44,000 speakers (INEGI 2020), although actual vitality is difficult to ascertain (Torres 2018). Typologically speaking, O’dam shares many properties with other highly agglutinating head-marking languages. The verb is the only obligatory clausal element and the relative ordering of larger phrases (DPs, PPs, CPs) is free, although the internal ordering of those phrases is rigid, except CPs. The frustratives we discuss here occur in the preverbal position (PreV), which otherwise consists of a range of scopally ordered clause-modifying particles common among V-initial languages (Carnie & Guilfoyle 2000). All of the O’dam data presented in this paper were collected via firsthand fieldwork by the first author.

Previous work has glossed both *tii* and *tiiip(up)* as marking “non-realized intention” (Willett 1991, Willett & Willett 2015). This reflects a shared counter-to-expectation effect (crosslinguistically characteristic of frustratives), but we will argue that the particles are not semantically equivalent. We begin by labeling them FRST1 and FRST2.

The first difference between the particles is illustrated by the minimal pair in (4). (4a) can be uttered by a speaker who has (unexpectedly) not yet mastered O’dam but is continuing to study and may still become fluent. *Tiiip(up)* cannot be used in this context: instead, while (4b) does convey that the speaker’s current grasp of O’dam is poor, it additionally rules out the possibility of future improvements, as indicated by the consultant comment. The near-minimal pair in (5) is similar: both examples convey that the bus has unexpectedly failed to arrive at reference time, but only (5a) can be used by a speaker who is still waiting.⁵ Informally, then, we find that while both particles target a particular event for frustration, only *tii* leaves open the possibility of a “better outcome” (i.e., that the frustrated event can still come to pass).

- (4) a. **tii** na=ñ machi-a’ gu o’dam
 FRST1 SUB=1SG.SBJ learn-IRR DET O’dam
 ‘I’m still learning O’dam’ (I will continue to learn it but currently I can’t speak it adequately)
 b. **tiiipup** na=ñ machi-a’ gu o’dam
 FRST2 SUB=1SG.SBJ learn-IRR DET O’dam

⁴ This language has also been called Southeastern Tepehuan. While that name does not appear to be viewed as derogatory, our consultants prefer the endonym O’dam. The name *tepehuan* is of Nahuatl origin, *tepe-wan* composed of *tepe-tl* ‘mountain’ + *-wan* ‘owners, dwellers’ likely meaning ‘mountain dwellers/owners,’ referring to where most Tepehuan peoples lived and continue to live.

⁵ Note that we use a near minimal pair in (5) because our consultant deemed (5b) unacceptable with irrealis marking on the verb.

‘I almost learned O’dam’ (but now I never will)

Comment: It’s like you got in a huge fight with all your consultants and are never coming back to Durango.

- (5) a. añ **tii** niira-’ gu camion
1SG.SBJ FRST1 wait-IRR DET bus
‘I’m waiting for the bus (but it still has not come)’ [said while you are waiting]
- b. añ **tiipup** niira-t gu camion
1SG.SBJ FRST2 wait-IMPF DET bus
‘I was waiting for the bus (but it never came)’

(4)-(5) also illustrate a contrast in temporal anchoring, despite O’dam lacking tense as a grammatical feature altogether: the examples using *tii* are compatible with present tense, while the *tiip(up)* examples are by default interpreted with past temporal reference.⁶ In general, we find that *tiip(up)* shows a strong preference for past interpretation, while *tii* is more flexible. We believe this difference is a result of the “better outcome” contrast: see §5.

Tense differences aside, the examples in (4)-(5) all have readings that fall under incomplete frustrativity as described in §2, showing that both O’dam particles are compatible with this function. Both particles can also be used averatively. In (6a), *tii* modifies a clause describing an event of buying chips at the store by indicating that the speaker turned around before arriving at the store. In keeping with the “better outcome” contrast described above, this example leaves open the possibility that the speaker will still go get the chips. (6b) is also averative: here, *tiip(up)* indicates that the speaker never even left for the store, despite making preparations to do so, and further conveys that there is now no chance of going.

- (6) a. cham bia’-iñ gu popotes, **tii** ba-ja-saba’n-mira-k-añi-ch mu tienda
NEG have-1SG.SBJ DET chips FRST1 CMP-3PL.PO-buy-MOV-PNCT-1SG.SBJ-PFV DIR store
‘I don’t have chips, I was going to buy them at the store’ (but I turned around)
- b. **tiipup** jii-ñi-ch mu tienda
FRST2 go.PFV-1SG.SBJ-PFV DIR store
‘I almost went to the store’ (but I never even left and now I won’t/can’t go)

Proper frustrativity is another matter: we only find these readings with *tii*. For instance, (7a) can convey that the subject reached La Candelaria (*koba’ram*), but the anticipated result (meeting Wendy) was not realized at reference time. (7b), however, can only be incomplete or averative. Our consultants find it acceptable if Michael was on his way but the bus broke down, or if he had plans to go but never departed. It cannot be used if he made it to La Candelaria: the parenthetical continuation in (7a) is rejected in (7b).

- (7) a. **tii** jii gu maikol koba’ram (dai na gu Wendy cham mu da-ka-t)
FRST1 go.PFV DET Michael La Candelaria (but SUB DET Wendy NEG DIR be.sitting-ST-IMPF)
‘Mike went to La Candelaria (but Wendy wasn’t there).’
- b. **tiipup** jii gu maikol koba’ram
FRST2 go.PFV DET Michael La Candelaria
‘Michael almost went to La Candelaria’ (but never left or the bus broke down on the way)

We find the same effect in (8): (8a) can express proper frustrativity, but *tiip(up)* is unacceptable if the snake has been killed. (8b) can only describe situations in which the snake remains alive: for instance, if the addressee only injured it, or if it escaped unharmed while the addressee was getting ready to strike.

- (8) a. ap **tii** mua dhi-ñi ko’
2SG.SBJ FRST1 kill.SG DEM.PROX-VIZ snake

⁶ A careful reader might observe that the verb is irrealis-marked in (5a) but imperfective in (5b), but we do not believe this is responsible for the tense contrast: imperfective marking in O’dam is not generally indicative of the past tense. Moreover, this explanation cannot account for the fully minimal pair in (4).

‘You killed this snake in vain’ (someone else took it to eat it)

- b. ap **tiipup** mua dhi’-ñi ko’
 2SG.SBJ FRST2 kill.SG DEM.PROX-VIZ snake
 Proper (unavailable): ‘#You successfully killed this snake, but . . .’
 Incompletive: ‘You almost killed this snake’ (but only hurt it)
 Avertive: ‘You were going to kill this snake’ (but it escaped)

The difference between proper frustrativity, on the one hand, and incompletive and avertive readings, on the other, is in whether or not an event in the denotation of the marked-clause predicate is (fully) realized: *tii* (in 7a and 8a) permits this, targeting a clause-external effect for frustration, but *tiip(up)* does not. The contrast is illustrated even more sharply by (9): with *tii*, this describes a situation in which the banana is still unripe (frustrating the expectation that it should by now be ripe), but our consultants outright reject *tiip(up)* here. Incompatibility with (full) predicate realization blocks a proper reading on which the banana is not yet ripe, and since it is neither possible for a banana to be in the developmental process of achieving the unripe state (incompletive), nor to exist in a state that developmentally or causally precedes unripeness (avertive), there is simply no coherent reading for *tiip(up)* in (9).

- (9) **tii** / **#tiipup** cham ba-x=bhai’ gu karum
 FRST1 / #FRST2 NEG CMP-COP=ripe DET banana
 ‘The banana is (still) not ripe’ (only *tii*)

As noted in §2, incompletive frustrativity can have a progressive-like character, typically associated with events that realize only the process phase of an accomplishment. O’dam accomplishments are compatible with incompletivity (as shown in 4-5, 7b, 8b), and we find that *tiip(up)* licenses only avertivity with punctual achievements (cf. 2c). (10) is an interesting example of this effect: the verb *mataimda*’ (‘nixtamalize’) describes the mixing of corn with a solution of lime and water to soften the kernels for making dough for tortillas. While the kernels require time to be ready for grinding, nixtamalization itself occurs as soon as the corn and solution are mixed, making *mataimda*’ an achievement. As a result, *tiip(up)* in (10b) can only have an avertive reading, while any events in which the results of nixtamalization were not as expected (e.g., the ground corn tastes wrong) must be described via proper frustrativity with *tii*.

- (10) a. **tii** tu-mataima’n-iñ dai na=ñi-ch gɛ’ mui’ bui’ñ matai
 FRST1 DUR-nixtamalize-1SG.SBJ but SUB=1SG.SBJ-PFV big DIR throw.PFV lime
 ‘I am nixtamalizing (corn) but I put too much lime in’
Comment: the corn will nixtamalize, but it will be different (lit. *fuerte* ‘strong’).
 b. **tiipup** na=ñi chu-mataimda-’
 FRST2 SUB=1SG.SBJ DUR-nixtamalize-IRR
 ‘I was going to nixtamalize (corn)’ (. . . but I didn’t because I already have enough tortillas)

Incompletive readings are usually not reported for frustrative clauses with atelic predicates, but we find a more nuanced picture in O’dam. Despite blocking proper frustrativity, *tiip(up)* is not restricted to describing developmentally incomplete portions of telic predicates: it can also combine non-avertively with a subset of atelic predicates. In (11), for instance, the speaker uses *tiip(up)* to indicate that they saw only part of Marcelo. The frustrative simultaneously precludes a full view while indicating that this was originally expected, and the comment in (11) highlights the effect by noting that the speaker must really like Marcelo to feel good in this scenario. By contrast, combining the atelic predicate in (12) with *tiip(up)* allows only avertive readings: our consultants reject this example if the dog was able to move at all.

- (11) xib **tiipup** tii-ñi-ch gu marcelo jix=bhai’ jiñ-chat-iñ
 today FRST2 see.PFV-1SG.SBJ-PFV DET Marcelo COP=good 1SG.MID-feel-1SG.SBJ
 ‘I got a glimpse of Marcelo today (e.g. through the grates of a fence), I feel great!’
Comment: It sounds like you’re a huge fan of Marcelo.

- (12) Context: some kids tied up the dog and then got it excited

ya' **tii**pup mii gu gagoox
 DIR FRST2 run.SG.PFV DET dog
 'The dog tried to run' (but could not move)

As far as we can tell, the difference between (11) and (12) lies in whether or not the underlying predicate can be validly realized while still describing an event that is *nonmaximal*. For instance, it is possible to see Marcelo while seeing only part of him: nonmaximality here is mediated through partial affectedness of the object, and the frustrative signals that while the instantiated event could in principle have become a maximal one, this did not happen. Nonmaximal instantiation is impossible in (12), however, since any amount of running constitutes a fully-realized running event; similarly, neither *tii* nor *tii*p(up) can license incompleteness in (9) because there is no way to achieve a partial state of unripeness that could plausibly become a maximal unripeness state (this is simply not how ripening works). Nonmaximality also appears to be the right characterization for *tii*-marked incompleteness. For instance, (13) can be used by a fluent speaker to signal that they do not know the O'dam term for a particular Spanish word: they know O'dam, but nonmaximally.⁷ Our conclusion is that the prohibition on 'proper' frustrativity reflects an *antimaximal* (rather than antiresultative) requirement imposed by *tii*p(up).

- (13) **tii** jix=maat-iñ gu ñi'oki-'ñ o'dam
 FRST1 COP=know-1SG.SBJ DET language-3SG.POSS O'dam
 'I know O'dam' (but not that word)

Our data show that O'dam frustratives differ in two key ways. First, *tii* encodes a 'weak' form of unexpectedness, indicating that the target of frustrativity is not (known to be) true at reference time, but may yet come to pass. *Tii*p(up) precludes the possibility of a "better" (non-frustrated) outcome. We believe these effects can be handled at the clause level, and align with existing treatments of frustrativity that capture the counter-to-expectation effect in terms of *inertial* modality. The other difference between *tii* and *tii*p(up) is subclausal: *tii* is compatible with both maximal and nonmaximal instantiations of an input predicate, but *tii*p(up) requires nonmaximal instantiation. We update our glosses to reflect this assessment, labeling *tii*p(up) as FRST.NONMAX and *tii* simply as FRST.

Finally, the surface realizations of the two particles suggest that there is a morphological relationship between them: specifically, that *tii*p(up) combines *tii* with a second morpheme, *pup*. Our descriptive generalizations are compatible with this view, and we develop a semantic analysis which treats *tii*p(up) as strictly stronger than *tii* (in keeping with Koontz-Garboden's 2007 *Monotonicity Hypothesis*). However, in the absence of clear and convincing evidence for the existence of a *pup* morpheme, we refrain from committing to a fully compositional treatment of *tii*p(up).⁸

4. Frustratives and aspect: prior approaches

Incompletive and avertive frustrativity diverge from proper frustrativity in whether the target of frustration is lexically or pragmatically specified. Viewed as an interaction between the frustrative and the event predicate, this variation appears idiosyncratic: while aspectual class can constrain the available readings, there does not seem to be a clear way to predict whether the frustrative target is clause-internal or external.

Copley & Harley (2014) suggest that a unified approach must instead treat frustratives propositionally. This is motivated by an apparent link between frustrativity type and grammatical aspect in Tohono O'dham: as examples (1c), (2a) and (3a) demonstrate, proper, incompletive and avertive readings appear to correlate (respectively) with the presence of perfective, imperfective and prospective (FUT) marking. Copley & Harley propose that frustrative *cem* asserts the proposition in its scope, leading to full realization of the marked-clause event in combination with perfective aspect, partial realization with the imperfective, and a preparatory reading with prospectivity. The frustrative itself further establishes the non-realization

⁷ *Tii*p(up) is rejected in this context, since it precludes improvement: one can always learn new words.

⁸ The existing evidence is inconclusive: Willett & Willett (2015: 147) define a particle *pup* as indicating the absence of intention, but our database contains only three independent examples of this morpheme, which we do not find sufficiently informative to warrant a clear conclusion. Our consultants have so far rejected *pup* outside of *tii*p(up).

of some expected consequence of the asserted situation, targeting some result of the completed event in (14a), the successful completion of this event in (14b), and its initiation in (14c), as desired. The proposal in (15) formalizes this approach in terms of inertial modality.⁹

- (14) a. **Proper:** (1c) $\equiv cem(\text{PFV}(\sqrt{\text{Juan open the door}}))$ *Juan opened the door in vain*
 b. **Incompletive:** (2a) $\equiv cem(\text{IMPF}(\sqrt{\text{Juan open the door}}))$ *Juan was trying to open the door*
 c. **Avertive:** (3a) $\equiv cem(\text{PROSP}(\sqrt{\text{Juan open the door}}))$ *Juan was going to open the door*

$$(15) \llbracket \text{FRST} \rrbracket := \lambda w \lambda p_{st}. p(w) \ \& \ \partial(\exists q_{st}(\neg q(w) \ \& \ \forall w' \in \text{INR}(w)[p(w') \rightarrow q(w')]))$$

Copley & Harley treat the non-inertial effect as presuppositional, citing data which shows that frustrativity outscopes negation in Tohono O’odham (see also Copley 2005); Davis & Matthewson (2022) come to the same conclusion for St’át’imcets. The effect is replicated in O’dam: (16a) indicates that the banana is ripe (but still cannot be eaten), and negation in (16b) targets only the ripeness predication. A reading where negation targets frustrativity—i.e., the banana is ripe and nothing unexpected follows—is impossible.¹⁰

- (16) a. **tii** ba-x=bhai’ gu karum b. **tii** cham ba-x=bhai’ gu karum
 FRST CMP-COP=ripe DET banana FRST NEG CMP-COP=ripe DET banana
 ‘The banana is ripe’ ‘The banana is (still) not ripe’
 Comment: It fell on the floor

Unfortunately, proposal (15) cannot apply directly to either *tii* or *tiiip(up)*. The non-inertial presupposition in is fairly specific, identifying a particular expected proposition which fails in the actual world. This is evidently too strong for *tii*. The presupposition is better suited to *tiiip(up)*, but in this case we lack a means for blocking proper frustrativity, predicted just in case **FRST** composes with a perfective clause.

We cannot solve the problem simply by stipulating that *tiiip(up)* is incompatible with perfectivity. Such a restriction seems neither explanatory nor accurate for O’dam (see 11 and 12). We have argued that incompletive frustrativity in O’dam is best characterized as (predicate-level) *nonmaximality*: a prohibition on perfectivity does not explain (for instance) why *tiiip(up)* can only combine avertively with the running predicate in (12). Moreover, it does not seem to be the case that *tiiip(up)* is incompatible with perfective aspect: we find perfective marking in combination with *tiiip(up)* in the avertive examples (6b), (7b) and (12) as well as in the incompletive (21a). This suggests—unsurprisingly, given the discussion in §3—that the mapping between aspect and frustrativity type in O’dam does not align with the predictions made by an approach like (15). We find a number of other ‘mismatches’ in the data: irrealis marking (tentatively linked to futurity) occurs in the avertive example (10b), but also in incompletive examples like (4a), (4b), and (5a), and incompletive readings are possible with both irrealis and imperfective marking (see 5).¹¹

These points lead to two conclusions, which we formalize in §5. First, the non-inertial (frustrative) effect of *tii* must be weaker than the presupposition in (15). Kroeger (2024) makes a similar suggestion for the Kimaragang frustrative *dara*, which patterns with *tii* in leaving open the ultimate status of its target. Second, the distribution of readings for O’dam frustratives mandates separating frustrativity type from grammatical aspect: we should instead tie the former to a broader event-based notion of (non)maximality.

⁹ The lexical entry in (15) is adapted from a similar proposal offered by Davis & Matthewson (2022: p.1358). Copley & Harley’s analysis is implemented in the *force dynamics* framework (Copley & Harley 2015), which treats dynamic events as deterministic maps between static situations: in this framework, the frustrative presupposition is one of *non-efficacy* applied to the topic situation, amounting to the claim that the results predicted by forces extant in the topic situation do not get realized. This is a fundamentally causal conception of frustrativity, which differs from the view put forward by Davis & Matthewson: the latter argue that it is enough for the actual proposition *p* to be expected to co-occur with the frustrated target *q*. Our O’dam data is so far consistent with the causal view, but we believe that variation of this type can be handled by manipulating the relevant modal flavour in a definition like (15).

¹⁰ Like *tii* and *tiiip(up)*, O’dam negation (*cham*) occurs in PreV. It seems to be obligatorily ordered after a frustrative, when the two co-occur: our consultants reject examples in which *cham* appears before *tii* or *tiiip(up)*.

¹¹ Aspectual marking does not seem to be obligatory, and we continue to investigate how it is expressed and/or inferred. While our glosses remain tentative, our consultants express sufficiently clear intuitions about the aspectual properties of various clauses that we feel confident that frustrativity does not neatly track grammatical aspect in O’dam.

5. O’dam frustrativity: non-inertia and antimaximality

We assume a branching time model (Thomason 1984), wherein the set of HISTORICAL alternatives to world w at time t contains worlds which are indistinguishable from w through t . The *inertial continuations* of a situation c at w, t are the accessible histories where c develops stereotypically (i.e., causally normally): following Kaufmann (2013) and Nadathur (2023), we assume that a CAUSAL ordering source can be constructed from a contextually-relevant *causal model* (Pearl 2000). This allows us to formulate two types of non-inertial commitment: (17b-i) expresses that w is generally atypical, while (17b-ii) identifies a propositional locus for atypicality. Finally, we assume that uninflected eventuality predicates have (potentially trivial) mereological structures, and define a function MAX that filters out nonmaximal events.

- (17) a. *Inertial continuations of c* : $\text{INR}(c, w, t) := \text{Best}_{\text{CAUS}(c, w, t)}(\cap \text{HIST}(w, t))$
 b. *Non-inertia*: Let w be a world, $c \subseteq w$ a situation, and t a time.
 i. w is *weakly non-inertial* at c iff $w \notin \text{INR}(c, w, t)$
 ii. w is *strongly non-inertial* at c iff $\exists q_{st} [\neg q(w) \ \& \ \forall w' \in \text{INR}(c, w, t) [q(w')]]$
 c. *Event maximality*: $\llbracket \text{MAX} \rrbracket := \lambda P_{vt} \lambda e. P(e) \ \& \ \forall e' [e \subseteq e' \rightarrow \neg P(e')]$

Both O’dam frustratives presuppose non-inertiality. To capture the “better outcome” contrast, we propose that *tii* encodes weak non-inertia, while *tiiip(up)* is stronger. Second, while frustrativity does interact with aspect, O’dam claims are sensitive to an event-based notion of maximality that is not wholly aspectually governed. As a result, we propose that *tii* and *tiiip(up)* scope above aspect, taking both an aspectual marker (perfective or imperfective) and an event predicate—which may in principle be maximally or nonmaximally realized—as arguments. The difference between proper, incomplete, and avertive readings is then mediated by (non)maximality: (i) proper frustrativity arises when the embedded predicate P is maximally instantiated, (ii) incomplete frustrativity arises with nonmaximal instantiation, and (iii) avertive frustrativity (following Kroeger 2024) is a special case of incomplete frustrativity, arising when P is coerced into a telic predicate of *inchoative* (causally-preparatory) events for P before inflection.¹² Per (18), *tii* permits both maximal and nonmaximal instantiations, and is therefore compatible with all three readings, but *tiiip(up)* encodes *antimaximality*, ruling out proper frustrativity.

- (18) Lexical entries for O’dam frustratives:
 a. $\llbracket \text{tii}(\text{ASP}, P) \rrbracket^{c, t} := \lambda w. \text{ASP}(P, w, t) \ \& \ \partial(w \notin \text{INR}(c, w, t))$
 b. $\llbracket \text{tiiipup}(\text{ASP}, P) \rrbracket^{c, t} := \lambda w. \text{ASP}(P, w, t) \ \& \ \partial(\exists q. \neg q(w, t) \ \& \ \forall w' \in \text{INR}(c, w, t) [q(w', t)])$
 $\ \& \ \partial(\neg \exists e [\text{MAX}(P)(e)(w) \ \& \ \tau(e) \circ t])$

A crucial assumption here is that both maximal and nonmaximal events can validly instantiate a predicate with non-trivial mereological structure. We likewise assume that O’dam aspects are indifferent to (non)maximality, so that their primary function is to situate events relative to worlds and times.

This raises questions about the relationship between nonmaximality and nonculmination. Standard theories of accomplishments assume a trivial internal structure, containing only culminated (thus, maximal) eventualities. While this approach is compatible with (18), provided we adopt an intensional theory of aspect (Dowty 1979, Altshuler 2014: a.o.), it means that incomplete frustrativity for accomplishments remains an aspectual phenomenon, conceptually distinct from atelic incompleteness. This is puzzling, as our consultants do not seem to regard the two types of incompleteness as fundamentally different. We propose, therefore, to adopt a recent proposal from Nadathur & Filip (2021), on which the denotation of a telic predicate P is imbued with a non-trivial internal structure via correspondence with a set of (context-sensitive) teleologically optimal alternatives for P ’s culmination condition. Space constraints prevent us from discussing the proposal in detail (see Nadathur & Everdell 2025), but its crucial consequence is that both culminated and non-culminated eventualities can validly instantiate telic predicates, making them subject to non-vacuous application of MAX, and aligning telic nonculmination with nonmaximality as desired. This allows us to combine (18) with a relatively simple treatment of grammatical aspects:

¹²A preliminary definition: $\llbracket \text{INCHO} \rrbracket := \lambda P_{vt} \lambda e. \exists e' [e \prec e' \ \& \ \text{CAUSE}(e, e') \ \& \ P(e')]$ (see Nadathur & Everdell 2025).

- (19) a. $\llbracket \text{PFV} \rrbracket := \lambda w \lambda t \lambda P_{vt}. \exists e [\tau(e) \subseteq t \ \& \ P(e)(w) \ \& \ \forall e' [e \sqsubseteq e' \rightarrow \neg P(e')(w)]]$
 b. $\llbracket \text{IMPF} \rrbracket := \lambda w \lambda t \lambda P. \exists e [\tau(e) \supset t \ \& \ P(e)(w)]$

One final puzzle remains. We based our definition of strong non-inertia on ideas articulated by Copley (2005) and Davis & Matthewson (2022): while it is aimed at capturing the definitive effect of *tiip(up)*—evidently stronger than *tii*—it is not clear to us that (17b-ii) accurately captures the desired effect. Work on incomplete and avertive frustrativity indicates a clear intuition that these readings target some or all of the input event predicate for frustration, but (17b-ii) only requires the frustrated target to be inertially linked to the evaluation context, and thus seems to be overly permissive. Since *tiip(up)* independently rules out maximal instantiations of the input predicate, this does not cause problems in (18b); however, the same observation suggests that the strongly frustrative effect of *tiip(up)* need not rely on strong inertia, but can instead be derived in the combination of antimaximality and weak inertia, resulting in the revised lexical entry in (20). Given (20), *tiip(up)* simply adds semantic content to *tii*, which is compatible with the existence of a (potentially diachronic) morphological relationship between the two (see §3).

$$(20) \llbracket \text{tiipup}(\text{ASP}, P) \rrbracket^{c,t} := \lambda w. \text{ASP}(P, w, t) \ \& \ \partial(w \notin \text{INR}(c, w, t) \ \& \ \neg \exists e [\text{MAX}(P)(e)(w) \ \& \ \tau(e) \circ t])$$

Our proposals derive the frustrativity contrasts discussed in §3. Weak inertia allows *tii* to avoid committing to the failure of a particular effect; antimaximality combined with weak frustrativity derives the strong and exclusively incomplete and avertive effects of *tiip(up)*. As suggested in §3, the contrast between (18a) and (20) predicts the difference in default temporal interpretation between *tii*- and *tiip(up)*-marked claims. To commit to event nonmaximality, a speaker needs access to information which determines how an instantiated event actually turned out: this is typically available only for past events. In contrast, (18a) only requires the speaker to have some (potentially predicate-independent) reason to believe that ongoing events will not turn out inertially, allowing a wider range of temporal anchors.¹³

Our analysis also provides some insight into the clause-scoping properties of O’dam frustratives. We find that *tii* and *tiip(up)* differ in the extent to which they can modulate the interpretation of clauses outside of the one in which they appear. For example, our consultants report that the minimally contrastive claims in (21) induce distinct interpretations for a subsequent copular clause, shown in brackets. Out of context, this clause asserts that the speaker is happy (lit. feels good). It maintains this interpretation in (21a). In (21b), however our consultants report a desiderative reading: the speaker seems to suggest that they want to (or should) feel happy (as a result of seeing Marcelo), but that this is not currently the case.

- (21) a. xib **tiipup** tii-ñi-ch gu marcelo [jix=bhai’ jiñ-chat-iñ]
 today FRST.NMAX see.PFV-1SG.SBJ-PFV DET Marcelo COP=good 1SG.MID-feel-1SG.SBJ
 ‘I got a glimpse of Marcelo today (e.g. through the grates of a fence), I feel great!’
Comment: It sounds like you’re a huge fan of Marcelo.
 b. xib **tii** tii-ñi-ch gu marcelo, [jix=bhai’ jiñ-chat-iñ]
 today FRST see.PFV-1SG.SBJ-PFV DET Marcelo COP=good 1SG.MID-feel-1SG.SBJ
 ‘I met Marcelo today, I (should) feel good’
Comment: You were sad or mad before meeting him and you were hoping he’d cheer you up.

We predict this pragmatic effect. Assuming that the speaker expected to see Marcelo properly, a nonmaximal sighting satisfies the conditions of non-inertiality with *tiipup*. *Tii*, however, allows a proper reading, indicating that the speaker saw Marcelo, but that events thereafter went awry: the juxtaposition of the copular clause raises the pragmatic likelihood this is the locus of non-inertiality. The effect of *tiip(up)* is clausally bound: informally speaking, it need not ‘frustrate’ events external to the marked clause. *Tii* has more freedom: since proper readings do not make the frustrative’s target explicit within the marked clause, frustrativity can affect any eventuality which is plausibly inertially related to the in-clause predicate.

¹³Our analysis also makes clear predictions about the interaction between *tii* and *tiip(up)* and a range of other frustrative functions (Overall 2017, Kroeger 2017), which we unfortunately cannot discuss here. Preliminary data is presented in Nadathur & Everdell (2025), but we leave detailed consideration of other frustrative uses for future work.

6. Conclusion

The analysis of frustrativity poses two challenges: explaining variation in how much of a marked event is realized and offering a consistent means of linking the marked clause to the frustrated outcome. Existing analyses link these puzzles to grammatical aspect, but—as we have seen—this is not enough to explain the distribution of readings for O'dam frustrative particles, which contrast both in terms of the (apparent) strength of their non-inertial commitments, as well as in the range of functions for which they can be used.

We proposed to handle the empirical landscape in O'dam by making a distinction between weak (non-specific) non-inertiality and strong non-inertiality: at least in O'dam, we argue that the latter effect arises in the combination of weak non-inertia and a predicate-level requirement of nonmaximality. Looking ahead then, our analysis suggests that non-inertial phenomena can be handled both above and below the clause level, and motivates reconsidering the role of event structure in apparently propositional counter-to-expectation effects. While a strong form of propositional non-inertiality may indeed be appropriate for frustratives in languages such as Tohono O'dham and St'át'imcets, our account opens up a range of typological possibilities, which may be adjudicated only by means of broader crosslinguistic study.

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